Course 502

Time 50 minutes   Marks 20

Answer any one

1. Consider a Ramsey-Cass-Koopmans model with the production function $Y = F(K, AL)$ and real interest rate r ; household's utility function takes the form

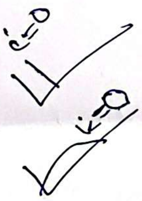
$$U=\int e^{-\rho t}u(c(t))\frac{L(t)}{H}dt$$

and there is CRRA(constant –relative –risk-aversion) utility

$$U(c(t))= \frac{C(t)^{1-\theta}}{1-\theta}$$

   a) Construct Household budget constraint and show that there is no-Ponzi-game condition. [5]
   b) Solve Euler equation for households' maximization problem. [7]
   c) Describe the behavior of the economy in terms of the evolution of c and k.[4]
   d) How does the competitive equilibrium maximize the welfare of the representative household. [4]

2. Suppose $Y_t = F(K_t, A_tL_t)$, with having constant returns to scale and the intensive form of the production function satisfying the Inada conditions. Suppose also that $A_{t+1} = (1+g)A_t$, $L_{t+1} = (1+n)L_t$ and the production function is Cobb–Douglas. Assume time is discrete rather than continuous and each individual lives for only two periods.
   a) Solve the individual's maximization problem where θ determines how much individuals varies in response to differences between r and ρ. [5]
   b) Find an expression for k_{t+1} as a function of k_t. [4]
   c) Describe how each of the following affects k_{t+1} as a function of k_t:
      (i) A rise in n.
      (ii) A rise in α. [3]
   d) Sketch k as a function of k_t. Does the economy have a balanced growth path? If the initial level of k differs from the value on the balanced growth path, does the economy converge to the balanced growth path? [4]
   e) Find the rate of convergence of k to k*. [4]



3. Consider a simplified version of the models of R&D and growth developed by P.Romer (1990),Gross and Helpman (1991) and Aghion and Howitt (1992) where θ reflects the effect of the existing stock of knowledge on the success of R&D.
   a) Explain the simplification process to construct the framework. [2]
   b) Critically explain the role of a_L (the labor force used in the R&D sector) for the dynamics of knowledge accumulation without considering capital into the model . [6]
   c) Assume that $\theta + \beta < 1$ and $n > 0$, and that the economy is on its balanced growth path. Describe how each of the following changes affects the $\dot{g}_A = 0$ and $\dot{g}_K = 0$ lines and the position of the economy in (g_A, g_K) space at the moment of the change:
      (i) An increase in n.
      (ii) An increase in a_K.
      (iii) An increase in θ. [6]
   d) Assume that $\beta + \theta = 1$ and $n = 0$,
      i) Find the value that A/K must have for g_K and g_A to be equal.
      ii) Find the growth rate of A and K when $g_K = g_A$.
      iii) How does an increase in s affect the long-run growth rate of the economy? [6]

competitive + complete
info present in RCK
no externality ⇒ welfare
maximized

Department of Economics
Jahangirnagar University
3rd Tutorial (M. Sc. Final Examination 2022)
Course Name: Health Economics
Course Number : Econ. 507
Full Marks: 20   Time: 45 Minutes

Answer any one(1) question:

1 [6+14]

a. What is Clinical Decision Analysis (CDA)? What are the steps of CDA?

b. Consider a patient who is 56 years old with "Heartburn" for several years. He may have coronary risk factors and a 6-week history of squeezing chest pain. In his **positive** physical examination physicians assigned **80%** coronary artery stenosis (prior probability). Sensitivity and specificity of the Exercise-ECG are **95% and 85%** respectively.

i) Calculate posterior probabilities using Bayes Theorem.

ii) Perform steps for CDA

iii) Graphically present sensitivity analysis based on two pre-test likelihoods.

2 [4+12+4]

a. Elucidate the concept of QALY as a measure of health benefits using Health Related Quality of Life (HRQoL) indicator.

b. Describe approaches in obtaining utility values for health states in Cost Utility Analysis (CUA)? Critically demonstrate Rating Scale (RS), Time-Trade Off (TTO) and Standard Gamble (SG) methods for assessing preferences of respondents.

c. Graphically explain relationship of cardinal utility with TTO method and SG method.

3 [20]

Briefly explain five-phase implementation of DHIS2 to transform the health information system in Bangladesh.

Eco 501 **Tutorial Exam** **Time: 2 hours** **Answer any 3**

1. a) What is preference based approach of consumer's behavior? Explain and interpret the assumptions of preference based approach.

   b) Define indifference set. Examine that if a preference relation satisfies strong monotonocity, its indifference curves must be downward sloping.

2. a) Define utility function. Explain the desirability of consumer's preference by the axiom of monotonocity and strong monotonocity.

   b) Show that,

$$StrongMonotonocity \underset{\not\Leftarrow}{\Rightarrow} Monotonocity \underset{\not\Leftarrow}{\Rightarrow} LNS$$

3. a) What are the twin definitions of convexity? In what respect they are different from strong convexity-examine graphically.

   b) Show that.

$$Convexity\ of\ preferences \Leftrightarrow USC(x)\,is\,convex \Leftrightarrow u(.)\,is\,quasiconcave$$

4. a) What are the typical utility functions used in economic applications? Write down their functional forms; also calculate and interpret the notions of MRS.

   b) Prove that, if a preference relation satisfies monotonocity and continuity, then there exist a utility function u(.) representing such preference relation.

5. a) Consider a utility maximization problem. If the utility function is continuous and preferences satisfy LNS over the consumption set $X = R_+^N$, then examine the properties satisfied by the Walrasian demand function $x(p,w)$. In what situations the sufficient condition of UMP may be violated.

   b) Examine the following properties of preference relations:

   i) Homogeneity

   ii) Homotheticity

6. Consider the Cobb-Douglas utility function:

$$u(x_1,x_2) = Kx_1^{\alpha}x_2^{1-\alpha} \quad \textit{for some } \alpha \in (0,1) \textit{ and } K > 0$$

i) Derive the Walrasian demand functions and examine their properties.

ii) How do you get the indirect utility functions? Explain their properties and also verify Roy's identity.

iii) Solve the expenditure minimization problem and examine the characteristics of the compensated demand function.

7. a) Distinguish between compensating variation (CV) and equivalent variation (EV). In what respect they are different from the consumer's surplus (CS)

b) Illustrate that for a i) decrease in price of a commodity (when both goods are normal)

$$EV > CS > CV$$

ii) for price increase $EV < CS < CV$

8. a) Define production sets. Graphically explain the properties of the production set.

b) What is the elasticity of substitution (EoS). Find the elasticity of substitution for the production function $q = f(K, L) = aK + bL$. What will be the EoS if the production function is fixed proportions?

9. a) Define profit function. If the production set Y is closed and have free disposal, then what properties do the profit function satisfies.

b) In case of well-defined utility function, show that $Concavity \rightleftarrows \!\!\!\!\!/\; Quasiconcavity$

10. Consider a consumer with the utility function: $u(x_1, x_2) = x_1^{\frac{1}{4}} x_2^{\frac{1}{2}}$

i) Derive the Walrasian and compensated demand functions.

ii) Consider a) $\{p^0, m^0\} = \{4, 4, 300\}$

b) $\{p^1, m^0\} = \{2, 4, 300\}$

Compute the indirect utilities and costs.

iii) Calculate EV, CV and CS. Based on their relationship examine the nature of the commodities.

Department of Economics

Jahangirnagar University

1st and 2nd Tutorial (M. Sc. Final Examination 2022)

Course Name: Health Economics

Course No. Econ. 507

Marks: 20+20 Time: 40 Minutes + 40 Minutes

Tutorial 1: Answer any one

1. *What is the definition of Health Economics?* Give a bird eye view of Health Economics as a separate *discipline. How are supply and demand tools* applied in the healthcare sector?
2. *What are the public interventions in health* sector that impacts private decisions? Explain. How cost *of health care services became major challenges* in the HC sector? How to overcome these *challenges?*

Tutorial 2: Answer any one

1. Why do some employers choose not to provide insurance, or find themselves unable to do so? How does Mandated Coverage exacerbate this phenomenon? working uninsured + mandated coverage.
2. How the tax system influences health insurance demand? Explain how health insurance may affect labour supply related to retirement age and job mobility?

Course 502

Time 50 minutes

Marks 20

1. (a) Show the impacts of saving ratio in the steady state solution of Solow growth model and prove that exogenous changes in the saving ratio leave only level effects on k and y, but not growth effects.
   (b) Show the impacts of changes in the saving ratio on per capita consumption and long run output. Does any steady state solution automatically imply golden rule of capital accumulation? Discuss. [6+6]

2. Find the elasticity of output per unit of effective labor on the balanced growth path, $y*$, with respect to the rate of population growth, n. If $\alpha_K(k*) = 1/3$, $g = 2\%$, and $\delta = 3\%$, by about how much does a fall in n from 2 percent to 1 percent raise $y*$? [4]

3. *Briefly* explain classical dichotomy and neutrality of money using the pillars of classical macroeconomics. [4]